

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv('/content/drive/MyDrive/TeachingRatings.csv')
```

```
df.head()
```

	rownames	minority	age	gender	credits	beauty	eval	division	native	tenure	students	allstudents	prof	
0	1	yes	36	female	more	0.289916	4.3	upper	yes	yes	24	43	1	
1	2	no	59	male	more	-0.737732	4.5	upper	yes	yes	17	20	2	
2	3	no	51	male	more	-0.571984	3.7	upper	yes	yes	55	55	3	
3	4	no	40	female	more	-0.677963	4.3	upper	yes	yes	40	46	4	
4	5	no	31	female	more	1.509794	4.4	upper	yes	yes	42	48	5	

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.shape
```

(463, 13)

```
df.columns
```

Index(['rownames', 'minority', 'age', 'gender', 'credits', 'beauty', 'eval',
 'division', 'native', 'tenure', 'students', 'allstudents', 'prof'],
 dtype='object')

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 463 entries, 0 to 462
Data columns (total 13 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   rownames    463 non-null   int64
1   minority    463 non-null   object
2   age         463 non-null   int64
3   gender      463 non-null   object
4   credits     463 non-null   object
5   beauty      463 non-null   float64
6   eval        463 non-null   float64
7   division    463 non-null   object
8   native      463 non-null   object
9   tenure      463 non-null   object
10  students    463 non-null   int64
11  allstudents 463 non-null   int64
12  prof        463 non-null   int64
dtypes: float64(2), int64(5), object(6)
memory usage: 47.2+ KB
```

```
df.describe()
```

	rownames	age	beauty	eval	students	allstudents	prof	
count	463.000000	463.000000	4.630000e+02	463.000000	463.000000	463.000000	463.000000	
mean	232.000000	48.365011	6.263499e-08	3.998272	36.624190	55.177106	45.434125	
std	133.800847	9.802742	7.886477e-01	0.554866	45.018481	75.072800	27.508902	
min	1.000000	29.000000	-1.450494e+00	2.100000	5.000000	8.000000	1.000000	
25%	116.500000	42.000000	-6.562689e-01	3.600000	15.000000	19.000000	20.000000	
50%	232.000000	48.000000	-6.801430e-02	4.000000	23.000000	29.000000	44.000000	
75%	347.500000	57.000000	5.456024e-01	4.400000	40.000000	60.000000	70.500000	
max	463.000000	73.000000	1.970023e+00	5.000000	380.000000	581.000000	94.000000	

Q1 — Time Series or Cross-Sectional?

Q1. Type of Data

The TeachingRatings dataset is cross-sectional data.

The dataset contains observations of 463 university courses/instructors. Although the observations were collected during the academic years 2000–2002, the same instructors are not systematically followed over time as a time series.

Therefore, the TeachingRatings dataset is classified as cross-sectional data.

Q2 — Mean, Median, Minimum and Maximum of Students

```
df['students'].head()
```

	students
0	24
1	17
2	55
3	40
4	42

dtype: int64

```
print("Mean:", df['students'].mean())
print("Median:", df['students'].median())
print("Minimum:", df['students'].min())
print("Maximum:", df['students'].max())
```

```
Mean: 36.62419006479482
Median: 23.0
Minimum: 5
Maximum: 380
```

Q3 — Descriptive Statistics Table

```
df.describe()
```

	rownames	age	beauty	eval	students	allstudents	prof
count	463.000000	463.000000	4.630000e+02	463.000000	463.000000	463.000000	463.000000
mean	232.000000	48.365011	6.263499e-08	3.998272	36.624190	55.177106	45.434125
std	133.800847	9.802742	7.886477e-01	0.554866	45.018481	75.072800	27.508902
min	1.000000	29.000000	-1.450494e+00	2.100000	5.000000	8.000000	1.000000
25%	116.500000	42.000000	-6.562689e-01	3.600000	15.000000	19.000000	20.000000
50%	232.000000	48.000000	-6.801430e-02	4.000000	23.000000	29.000000	44.000000
75%	347.500000	57.000000	5.456024e-01	4.400000	40.000000	60.000000	70.500000
max	463.000000	73.000000	1.970023e+00	5.000000	380.000000	581.000000	94.000000

Q4 — Histogram of Beauty

```
df['beauty'].describe()
```

	beauty
count	4.630000e+02
mean	6.263499e-08
std	7.886477e-01
min	-1.450494e+00
25%	-6.562689e-01
50%	-6.801430e-02
75%	5.456024e-01
max	1.970023e+00

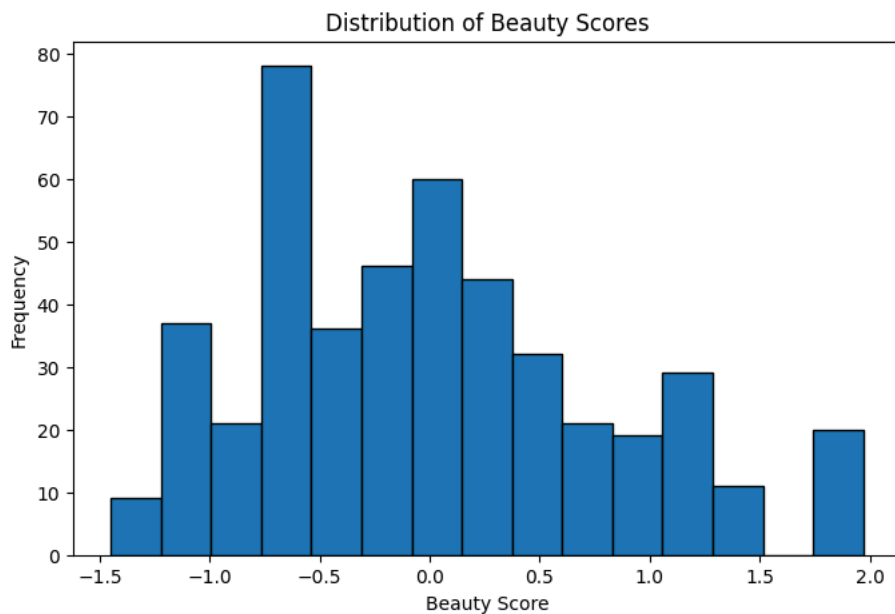
dtype: float64

```
plt.figure(figsize=(8,5))

plt.hist(df['beauty'], bins=15, edgecolor='black')

plt.xlabel('Beauty Score')
plt.ylabel('Frequency')
plt.title('Distribution of Beauty Scores')

plt.show()
```



df['beauty'].skew()

np.float64(0.5158042718205115)

Q5 — Does Average Beauty Score Differ by Gender?

df['gender'].value_counts()

	count
gender	
male	268
female	195

dtype: int64

df.groupby('gender')['beauty'].mean()

```

        beauty
gender
female  0.116109
male    -0.084482

dtype: float64

```

```
df.groupby('gender')['beauty'].agg(['count', 'mean', 'std'])
```

```

      count    mean    std
gender
female   195  0.116109  0.81781
male    268 -0.084482  0.75713

```

```

gender_means = df.groupby('gender')['beauty'].mean()

difference = gender_means['female'] - gender_means['male']

print("Difference in mean beauty score:", difference)

```

```
Difference in mean beauty score: 0.20059131148679676
```

```

from scipy.stats import ttest_ind

male = df[df['gender'] == 'male']['beauty']
female = df[df['gender'] == 'female']['beauty']

t_stat, p_value = ttest_ind(
    male,
    female,
    equal_var=False
)

print("t-statistic:", t_stat)
print("p-value:", p_value)

```

```

t-statistic: -2.6880162950066833
p-value: 0.007488930799966839

```

Q6 — Percentage of Males and Females Who Are Tenured

```
df['tenure'].value_counts()
```

```

      count
tenure
yes       361
no        102

dtype: int64

```

```
pd.crosstab(df['gender'], df['tenure'])
```

```

tenure no  yes
gender
female  50  145
male    52  216

```

```

tenure_gender_percent = pd.crosstab(
    df['gender'],
    df['tenure'],
    normalize='index'
) * 100

```

```
tenure_gender_percent.round(2)
```

tenure	no	yes
gender		
female	25.64	74.36
male	19.40	80.60

```
from scipy.stats import chi2_contingency

table = pd.crosstab(df['gender'], df['tenure'])

chi2, p, dof, expected = chi2_contingency(table)

print("Chi-square statistic:", chi2)
print("p-value:", p)
print("Degrees of freedom:", dof)
```

```
Chi-square statistic: 2.20678166999886
p-value: 0.1374050603563787
Degrees of freedom: 1
```